

When Does Better Conditioning Help? Representation, Injection, and Sampling-Time Guidance in Latent Diffusion Super-Resolution

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Abstract

Latent diffusion super-resolution (SR) conditions a reverse process on a low-resolution (LR) latent z_{lr} . A natural hypothesis is that a better z_{lr} should yield better SR. We test that hypothesis under a controlled $4\times$ CelebA protocol ($32 \rightarrow 128$) with matched denoisers, frozen autoencoders, and per-image reverse-process noise. An SR-aware VAE improves the LR code substantially: soft-decode $D(z_{\text{lr}})$ gains $+2.33$ dB PSNR and LPIPS $0.273 \rightarrow 0.119$ ($n=64$), while HR reconstruction stays at ≈ 37.3 dB. The same gain does *not* transfer through a concat latent DDPM: SR PSNR moves only $+0.22$ dB and LPIPS is a null (≈ 0.068); imagewise Spearman $\rho(\Delta_{\text{soft}}, \Delta_{\text{SR}}) \approx 0.05$. Replacing concat with spatial FiLM is also a null (PSNR $\Delta = -0.03$ dB, CI includes 0). A timestep diagnostic localizes the failure: the SR-aware model becomes strongly aligned with z_{lr} mid-reverse (cosine 0.985 at $t=656$) and subsequently loses this alignment as sampling approaches the final HR latent. Inference-time decoder-side LR-consistency guidance, applied in the late sampling window $t \leq 500$ (no training), recovers **56%** of the 2.08 dB transfer gap at $\lambda_g=200$ ($n=256$: 26.33 $\rightarrow$ 27.50 dB) and slightly *improves* LPIPS relative to the unguided baseline (0.0720 $\rightarrow$ 0.0707). Our contribution is a controlled diagnosis of conditioning transfer in latent diffusion SR and an inference-time intervention that recovers a substantial fraction of the lost fidelity.

1 Introduction

Generative super-resolution (SR) increasingly runs in a *latent* space: an autoencoder compresses images, a diffusion model denoises the high-resolution (HR) latent, and a decoder returns pixels [Rombach et al., 2022, Saharia et al., 2023, Wang et al., 2024]. The low-resolution (LR) observation enters as a condition, typically the encoded upscaled LR image $z_{\text{lr}} = E(x_{\text{lr}}^\uparrow)$. This design assumes a convenient

causal story: if z_{lr} is a better representation of the HR target, the reverse process should emit a better z_{sr} , and SR metrics should follow.

That story is not a theorem. It confuses two jobs that share a decoder but not an objective (Table 1): soft-decode $D(z_{\text{lr}})$ scores what is already in the LR code; LATENTSR $D(z_{\text{sr}})$ scores what a conditional DDPM uses while sampling $p(z_{\text{hr}} | z_{\text{lr}})$.

This paper is a controlled diagnosis of that transfer, not a new SOTA SR method. We freeze the data, the UNet skeleton, the noise schedule, and the evaluation noise, and change one link at a time (Figure 1): representation (VAE-SR), injection (concat vs. spatial FiLM), reverse-chain measurement, then inference-time LR-consistency guidance. Headline outcomes—a large soft-decode gain that concat and FiLM do not consume, a mid-reverse copy that later collapses, and late guidance that recovers about half the gap—are in §5. The intervention that works is not the one the representation hypothesis suggested.

Scope. No new backbone, tokenizer, or trained guidance network. The FiLM null is for this conditioner on this frozen VAE-SR code. $D(z_{\text{lr}})$ is not an SR method. Citation hygiene: §2.5.

2 Problem setup and pipeline

This section fixes notation, the two decode jobs, the data and denoiser, and the evaluation protocol used by every later table. Method variants (VAE-1 vs. VAE-SR, concat vs. FiLM, guidance) are defined in §2–§2.5 only as *interfaces*; training losses and ablations come in subsequent sections.

2.1 Task and notation

Let $x_{\text{hr}} \in \mathbb{R}^{3 \times 128 \times 128}$ be an HR RGB face and $x_{\text{lr}} \in \mathbb{R}^{3 \times 32 \times 32}$ its $4\times$ bicubic downsample. Write $x_{\text{lr}}^\uparrow$ for bicubic upsample of x_{lr} back to 128×128 . A convolutional VAE provides an encoder–decoder pair (E, D) with spatial latent shape $(4, 32, 32)$ (downsample factor 4). The encoder outputs a diagonal Gaussian $q(z | x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x)))$. At SR inference we freeze the VAE

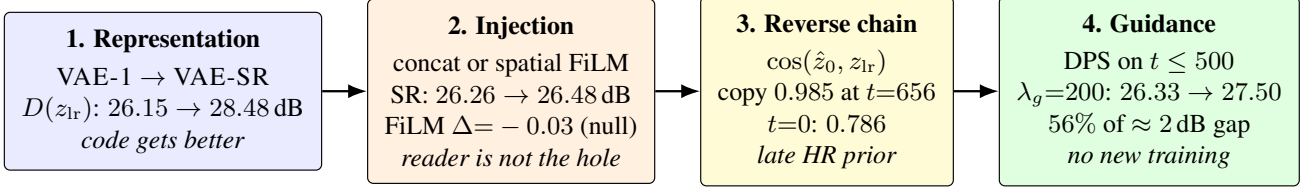


Figure 1: Causal chain. Each box changes one link and holds the others fixed. Numbers are the locked outcomes of §5 ($n=64$ except the last box, which is the $n=256$ dose).

and use the posterior *mean*

$$z = \mu(x), \quad z_{\text{scaled}} = sz, \quad D_{\text{scaled}}(z) = D(z/s), \quad (1)$$

with $s=1$ throughout (a scale $1/\text{std}(\mu_{\text{hr}}) \approx 1.065$ was estimated and *not* applied).

The diffusion model is a DDPM [Ho et al., 2020] on the HR latent $z_{\text{hr}} = E(x_{\text{hr}})$, with $T=1000$ steps and a linear schedule $\beta_t \in [10^{-4}, 0.02]$. We write $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ in the usual way. The UNet predicts noise $\hat{\epsilon}_\theta(z_t, z_{\text{lr}}, t)$. The predicted clean latent is

$$\hat{z}_0(t) = \frac{z_t - \sqrt{1 - \bar{\alpha}_t} \hat{\epsilon}_\theta(z_t, z_{\text{lr}}, t)}{\sqrt{\bar{\alpha}_t}}. \quad (2)$$

Reverse sampling uses the standard DDPM posterior step $p(z_{t-1} | z_t, \hat{\epsilon})$. The SR output is $z_{\text{sr}} \approx z_0$ after $t=0$, decoded as $D(z_{\text{sr}})$.

Two decodes. Table 1 is the wording constraint for the rest of the paper. Soft-decode reconstructs the *condition*. LATENTSR samples an HR latent. We never say that the DDPM “overwrites a better reconstruction.”

Table 1: Two jobs that share D but not an objective.

Name	Definition	Question
Soft-decode	$D(z_{\text{lr}}) = D(E(x_{\text{lr}}^\dagger))$	What HR-looking information is already in z_{lr} ?
LATENTSR	$D(z_{\text{sr}})$, $z_{\text{sr}} \sim p(z_{\text{hr}} z_{\text{lr}})$	How much of that information does the sampler use?

2.2 Data, architecture, and training constants

Table 2 lists the constants that never change across the causal chain. CelebA faces keep the problem in a well-understood, aligned domain; the point is isolation, not real-world degradation modeling. The denoiser is the project’s own latent UNet (`generative_models.ddpm`), not a pretrained Stable Diffusion backbone, so representation, injection, and sampling effects are not confounded with a frozen billion-parameter prior.

Table 2: Locked experimental constants. Changing any row without a paired re-eval invalidates a comparison.

Item	Value
Dataset	CelebA, center-crop 178, then resize
HR / LR	$128 \times 128 / 32 \times 32$ (4× bicubic)
Valid split	official; eval in file order, <code>shuffle=False</code>
Valid size	$\approx 19,867$ (full val never run)
Seed	42; x_T and every reverse-step noise seeded per <code>val_index</code>
VAE	conv. β -VAE, base channels 64, <code>channel_mult=[1, 2, 4]</code> , 2 resblocks
Latent	$(4, 32, 32)$, $s=1$
DDPM	$T=1000$, $\beta \in [10^{-4}, 0.02]$, Adam 2×10^{-4}
UNet	same skeleton as the VAE width, attn. at 16 and 8
Concat DDPM	50 epochs, batch 64

2.3 Pipeline

Figure 2 is the system at inference. HR and LR share one frozen encoder. The denoiser never sees pixels: it sees z_t and z_{lr} only. Conditioning is either channel concat

$$\hat{\epsilon}_\theta(z_t, z_{\text{lr}}, t) = \text{UNet}([z_t \| z_{\text{lr}}], t) \quad (3)$$

(8 input channels, 4 output channels) or spatial FiLM at every UNet scale (defined when that experiment is introduced). Guidance, when used, does not change (3): it adds a decoder-side correction after the unguided posterior step (§4.6).

2.4 Evaluation protocol

Every paired claim uses the same N CelebA validation images in file order, seed 42, and reverse-process noise drawn as a function of `val_index`. Two checkpoints are therefore comparable even if batch size differs. Primary metrics are PSNR (distortion) and LPIPS-Alex [Zhang et al., 2018] (perception). SSIM and Sobel edge MAE are secondary. Paired tests: Δ is candidate minus baseline on matched indices; sign-flip permutation $n_{\text{perm}}=10^4$; percentile bootstrap 95% CI $n_{\text{boot}}=10^4$. We declare a mean Δ real when the CI excludes 0.

Number hygiene. VAE, concat, FiLM, timestep, and $\hat{z}_0$ -image diagnostics are reported at **n=64**. The guidance

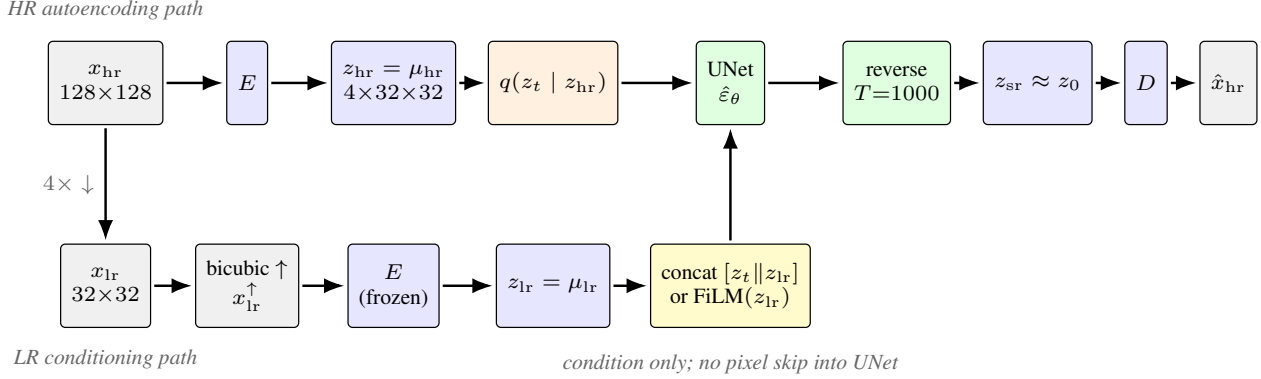


Figure 2: LatentSR inference pipeline (every stage that later tables refer to). Blue: frozen VAE. Orange: forward diffusion on z_{hr} (training) / noise at $t=T$ (eval). Green: concat or FiLM denoiser. Yellow: the only place the LR code enters the UNet. Soft-decode is $D(z_{lr})$ and never uses the green path. LATENTSR is $D(z_{sr})$.

dose that we treat as confirmatory is **n=256** (val_index 0..255), pre-registered before looking at those numbers. We never pool the two n in one mean. Full validation ($\approx 19,867$) is not computationally feasible at ≈ 21 s/image for guided reverse sampling.

2.5 Checkpoint pairing and hygiene

Three frozen VAEs / DDPMs are easy to mix because several artifacts are named `latest.pt`. Table 3 is the pairing used in every later section. Guidance is always Q2 concat epoch-50 *with* VAE-SR epoch-20. Phase-8 concat (trained on VAE-1) and the FiLM checkpoint are never used for guidance.

Table 3: Checkpoint pairing used in every later section.

Name	Frozen VAE	Role
VAE-1	recon. VAE, epoch 50	representation baseline
VAE-SR	20-epoch align fine-tune	representation test
Phase-8 concat	VAE-1	not for guidance
Q2 concat	VAE-SR	transfer + guidance pair
FiLM Q2	VAE-SR	injection test only

Citation hygiene. On 2026-08-19 a related-work list named a paper as “PGSR — When Latents Forget Pixels (2026)” without authors, arXiv ID, venue, or DOI. The acronym collides with planar-based Gaussian splatting; a circulating code URL did not check out as that SR paper. We exclude it: a title-plus-year string is not a citation.

3 Related work

We group prior work by the three links we ablate—how z_{lr} is *represented*, how it is *injected* into a denoiser, and how

sampling can be *steered* back toward the LR observation—and then state what is *not* claimed.

3.1 Latent diffusion and generative SR

Denoising diffusion [Ho et al., 2020] produces high-quality images by reversing a Gaussian noising process. Latent diffusion [Rombach et al., 2022] moves that process into the code of a convolutional VAE [Kingma and Welling, 2014], which is the template for LATENTSR: encode, denoise, decode. Pixel-space iterative SR [Saharia et al., 2023] showed that a conditional DDPM can outperform regression on perceptual metrics; subsequent systems adapt large pretrained latent priors for restoration, including StableSR [Wang et al., 2024], DiffBIR [Lin et al., 2024], and SeeSR [Wu et al., 2024]. Those methods are engineering stacks (ControlNet branches, semantic prompts, degradation-aware encoders) on frozen billion-parameter backbones. Our denoiser is a small latent UNet trained from scratch on CelebA [Liu et al., 2015], on purpose: representation, injection, and sampling can be swapped without confounding a pretrained generative prior.

The perception–distortion trade-off [Blau and Michaeli, 2018] is the right backdrop for every SR table below. LPIPS [Zhang et al., 2018] tracks the perceptual axis; PSNR tracks distortion. A method can win LPIPS while losing PSNR, and a soft-decode that looks closer to HR than the sampler’s output is not a bug in the metrics.

3.2 Representations for restoration

A reconstruction VAE is optimized for $D(z_{hr}) \approx x_{hr}$, not for z_{lr} to be a useful SR condition. Related restoration work often adds pixel-space skip connections, second encoders, or pretrained tokenizers. We stay inside one frozen VAE so that “better z_{lr} ” is a measured change in $D(z_{lr})$ and $\cos(z_{lr}, z_{hr})$ —not a new architecture. The hypothesis we test is the one implicit in much latent SR: if those

representation metrics improve, SR metrics should follow.

3.3 How the LR code is injected

The default in latent SR, and our baseline, is channel concat $[z_t \| z_{lr}]$ at the UNet input [Rombach et al., 2022]. Concat is spatially aligned—our z_{lr} is already $4 \times 32 \times 32$ —but it touches only the first convolution. Feature-wise modulation (FiLM; Perez et al., 2018) and its normalization cousins (AdaIN/AdaGN) inject a condition as γ, β at many depths, the natural test of “concat cannot read z_{lr} after the first conv.” ControlNet-style side branches [Zhang et al., 2023] and cross-attention are the other common readers. We use neither: cross-attention would treat an already-aligned map as tokens, and ControlNet would add a second encoder. A FiLM null on the same frozen z_{lr} is evidence against the weak-reader story, not a claim that FiLM never helps in other stacks.

3.4 Sampling-time guidance and prior vs. fidelity

Classifier guidance [Dhariwal and Nichol, 2021] and classifier-free guidance [Ho and Salimans, 2022] steer a reverse process with a gradient or a conditional/unconditional pair. Diffusion posterior sampling (DPS; Chung et al., 2023) applies a measurement gradient through a differentiable forward model, typically to z_{t-1} after an unguided posterior step. SeeSR [Wu et al., 2024] and DiffBIR [Lin et al., 2024] discuss generative priors that invent plausible but unfaithful detail; they add semantic or restoration conditions during sampling or via extra trained modules. Our probe uses that DPS pattern with an LR-consistency measurement through a frozen decoder, in a pre-registered time window, with λ_g chosen from a relative step size $R(t)$ rather than from a test-set sweep disguised as a method (§4.6).

3.5 Positioning

The experiment is narrow: hold the denoiser family and evaluation noise fixed, improve z_{lr} by a known amount, and ask whether concat, then FiLM, then late-window guidance consume that gain. We do not claim a new conditioning architecture or reproduce large-prior pipelines. A claimed related paper could not be verified from the citation as given and is excluded (§2.5).

4 Method

We change one link of Figure 2 at a time. This section specifies the four interventions—representation, injection, trajectory measurement, and sampling-time guidance—before any SR table. Training hyperparameters that never change are in Table 2; checkpoint pairing is in Table 3.

4.1 Stage A: reconstruction VAE (VAE-1)

VAE-1 is a convolutional β -VAE [Kingma and Welling, 2014] trained only to reconstruct HR CelebA faces. It has no SR term. The encoder outputs $(\mu, \log \sigma^2)$; training uses

$z = \mu + \sigma \odot \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$. The loss is mean-reduced MSE plus a mean-reduced KL, so the KL scale stays stable at latent size $4 \times 32 \times 32$:

$$\mathcal{L}_{\text{VAE-1}} = \underbrace{\|D(z) - x_{hr}\|_2^2}_{\mathcal{L}_{\text{recon}}} + \underbrace{\beta \mathbb{E}\left[-\frac{1}{2}(1 + \log \sigma^2 - \mu^2 - \sigma^2)\right]}_{\text{KL}}, \quad (4)$$

with $\beta = 10^{-4}$ and $\text{KL} = \text{KL}(q(z | x_{hr}) \| \mathcal{N}(0, I))$. We train for 50 epochs (Adam 10^{-4} , batch 64) and freeze epoch 50 for every later experiment. Eval reconstructions use μ , not a posterior sample.

4.2 Stage B: SR-aware alignment (VAE-SR)

VAE-SR is VAE-1 fine-tuned so the LR encoder mean moves toward the HR encoder mean without rebuilding the autoencoder (Figure 3). Initialize from VAE-1 epoch 50. On each batch encode HR and the bicubic-upsampled LR with the *same* encoder

$$\mu_{hr}, \log \sigma_{hr}^2 = E(x_{hr}), \quad \mu_{lr} = E(x_{lr}^\uparrow), \quad (5)$$

and optimize

$$\begin{aligned} \mathcal{L}_{\text{VAE-SR}} = & \mathcal{L}_{\text{recon}}(x_{hr}, D(z_{hr})) \\ & + \beta \text{KL}(q(z | x_{hr}) \| \mathcal{N}(0, I)) \\ & + \lambda_{\text{align}} \|\mu_{lr} - \text{sg}(\mu_{hr})\|_2^2. \end{aligned} \quad (6)$$

Stop-gradient $\text{sg}(\cdot)$ on μ_{hr} is required: without it the HR code can collapse toward the blurry LR code; with it, only the LR mean is pulled. β is unchanged (10^{-4}). $\lambda_{\text{align}} = 10^{-3}$ because unweighted align MSE starts near 0.58; 10^{-3} keeps the term on the order of recon MSE. We fine-tune for 20 epochs (Adam 10^{-4} , batch 32) and freeze the result. A λ_{align} sweep is *not* part of the method: cosine(z_{lr}, z_{hr}) plateaus after epoch 1, so the next causal question is consumption, not more alignment.

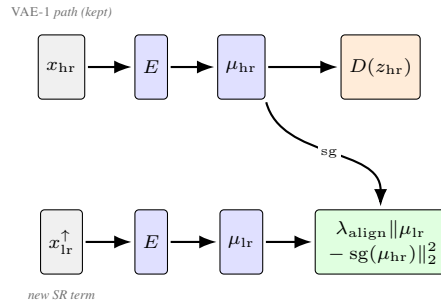


Figure 3: VAE-SR fine-tune. HR recon+KL is the VAE-1 loss (4). Alignment pulls μ_{lr} toward a stop-grad μ_{hr} . Soft-decode later scores $D(\mu_{lr})$, not this training grid.

4.3 Stage C: concat latent DDPM

Given a frozen VAE, we train a noise-prediction UNet on z_{hr} with channel-concat conditioning (3): eight input channels, four output channels, same skeleton and 50-epoch recipe for both VAEs. The training loss is standard DDPM noise MSE [Ho et al., 2020]. Only the frozen VAE changes between the VAE-1 concat run (Phase 8) and the VAE-SR concat run (Q2). We do not resume a concat checkpoint into FiLM (channel layout differs).

4.4 Stage D: spatial FiLM

To test whether concat is a weak reader of an already-better z_{lr} , we replace (3) with spatial FiLM [Perez et al., 2018] at every UNet scale. The noisy latent stays four channels. At a feature map h of matching (or bilinearly resized) resolution,

$$h' = (1 + \gamma(z_{\text{lr}})) \odot h + \beta(z_{\text{lr}}), \quad (7)$$

where γ, β come from a 1×1 convolution that is *zero-initialized*, so the block is identity at step 0. Cross-attention is not used (§3).

One-sided 2×2 . A full interaction would train FiLM on both VAEs. With one new DDPM, we train FiLM+VAE-SR and interpret it as a one-sided test, written down *before* seeing numbers: if SR still matches concat+VAE-SR, skip FiLM+VAE-1 and treat concat-as-reader as not the bottleneck; if FiLM+VAE-SR clearly beats concat+VAE-SR, then train FiLM+VAE-1 before claiming an interaction. Same 50 epochs, same paired eval as Q2.

4.5 Stage E: reverse-chain diagnostics

To localize where a better z_{lr} is or is not used, we record, at every scheduler index t , $\cos(\hat{z}_0(t), z_{\text{lr}})$ and $\|\hat{z}_0(t) - z_{\text{lr}}\|$ under both frozen VAEs and their matched concat DDPMs, plus the image-space scores of $D(\hat{z}_0(t))$ versus x_{hr} . Soft-decode $D(z_{\text{lr}})$ is not $D(\hat{z}_0(t))$ at a copy peak; those are reported separately.

4.6 Stage F: DPS-style LR-consistency guidance

Guidance is inference only: frozen VAE-SR + Q2 concat, no extra training. The convention is DPS-like [Chung et al., 2023] and is fixed before any guided SR number: gradient with respect to z_t , correction applied to the already-sampled z_{t-1} .

At each reverse index t :

1. $\hat{\varepsilon} = \text{UNet}(z_t, z_{\text{lr}}, t)$, *detached* (no UNet backward).
2. $\hat{z}_0$ as in (2).
3. $z_{t-1} \sim p(z_{t-1} | z_t, \hat{\varepsilon})$ (unguided DDPM step).
4. If t is in the window,

$$\begin{aligned} \mathcal{L}_g &= \text{mean}_{c,h,w} \|D(\hat{z}_0) - D(z_{\text{lr}})\|_2^2, \\ g &= \nabla_{z_t} \mathcal{L}_g, \\ z_{t-1} &\leftarrow z_{t-1} - \lambda_g g. \end{aligned} \quad (8)$$

$D(z_{\text{lr}})$ is cached once per image. Decoder weights stay frozen; the backward runs through the $\hat{z}_0$ affine and D only. Windows (scheduler index; $t=0$ is clean): baseline = none; early = $t \in [0, 800]$ every step; late = $t \in [0, 500]$ every step. A budget skip `guide_every > 1` is implemented but not used in the reported runs.

Choosing λ_g from $R(t)$, not from SR PSNR. Define the relative step size

$$R(t) = \frac{\|g\|_2}{\|z_{t-1} - z_t\|_2}. \quad (9)$$

A sanity sweep at $\lambda_g \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$ gives $R(t) < 0.01$ everywhere: $\lambda_g=1$ is a no-op. R is linear in λ_g . Table 4 scales the $\lambda_g=1$ column. $\|g\|/\|\Delta z\|$ drops by about $60\times$ from $t=800$ to $t=500$, so *one scalar cannot put every checkpoint in a 0.1–0.5 band*. We take $\lambda_g=200$ as a compromise: in-band at $t=650$ ($R \approx 0.18$); above 0.5 but below a 2–3 dominate ceiling at $t=800$ ($R \approx 1.16$); small but non-zero in the late window ($R(500) \approx 0.04$). $\lambda_g=400$ would make the late window fairer but $R(800) \approx 2.3$. A late PSNR-null is not treated as localization failure unless $R(t)$ for $t \leq 500$ is clearly > 0 .

Table 4: Relative guidance size $R(t)$ (9), scaled from a $\lambda_g=1$ measurement. Bold: operating column.

t	$R(1)$	$R(100)$	$R(200)$	$R(400)$
800	0.0058	0.58	1.16	2.32
650	0.0009	0.09	0.18	0.36
500	0.0002	0.02	0.04	0.08
300	0.0001	0.01	0.02	0.04
100	0.0001	0.01	0.02	0.04

Pre-registered calls (before guided PSNR). PSNR $\uparrow$ with LPIPS ≈ 0.068 is a strong positive; PSNR $\uparrow$ with LPIPS toward 0.119 is over-guidance. If both windows work, Stage 2 is a late-window dose $\lambda_g \in \{50, 200, 800\}$ only—no combined schedule. The $n=256$ confirmation (val_index 0..255, same checkpoints, late window) is declared before looking at those numbers, and survives iff (a) $\geq 90\%$ of images have $\Delta\text{PSNR} > 0$ at each λ_g , (b) mean ΔLPIPS is monotone $50 < 200 < 800$ with $\lambda_g=800$ worse than unguided and $\lambda_g=50$ not worse, (c) PSNR CIs exclude 0 at all three λ_g . The first line of each job must reprint PSNR($D(z_{\text{lr}}), x_{\text{hr}}$) ≈ 28.48 on indices 0..63; otherwise the wrong VAE is loaded and the run is discarded.

5 Experiments

Every table uses §2.4: same val_index order, seed 42, reverse-process noise as a function of index, pairing from Table 3. VAE, concat, FiLM, and trajectory are $n=64$; the confirmatory guidance dose is $n=256$. The two n are never pooled.

5.1 Representation

Freeze each VAE and score reconstructions with no diffusion (Table 5). Under VAE-1, $D(z_{lr})$ is essentially bicubic, while $D(z_{hr})$ is 37.3 dB—well above concat LATENTSR (≈ 26.3 dB). HR autoencoding is not the SR bottleneck. VAE-SR alignment lifts soft-decode +2.33 dB and LPIPS 0.273 $\rightarrow$ 0.119; cosine(z_{lr}, z_{hr}) 0.627 $\rightarrow$ 0.790; latent MSE 0.584 $\rightarrow$ 0.327. HR recon does not move: the align term did not collapse the autoencoder. A suggested scale $s = 1/\text{std}(\mu_{hr}) \approx 1.065$ is not applied (§2.1).

Table 5: VAE-only eval, $n=64$, no diffusion.

Method	PSNR $\uparrow$	LPIPS $\downarrow$	cos(z_{lr}, z_{hr})
Bicubic $x_{lr}^\uparrow$	26.13	0.282	—
VAE-1 $D(z_{lr})$	26.15	0.273	0.627
VAE-SR $D(z_{lr})$	28.48	0.119	0.790
VAE-1 $D(z_{hr})$	37.27	0.023	—
VAE-SR $D(z_{hr})$	37.28	0.024	—

Retrain the same concat DDPM for 50 epochs on each frozen VAE (Table 6). Soft-decode gained +2.33 dB; concat SR gained +0.22 dB. SR LPIPS is a tie at ≈ 0.068 . VAE-SR $D(z_{lr})$ (28.48 dB) is above concat SR (26.48 dB): the sampler did not use the extra 2 dB (Figure 4).

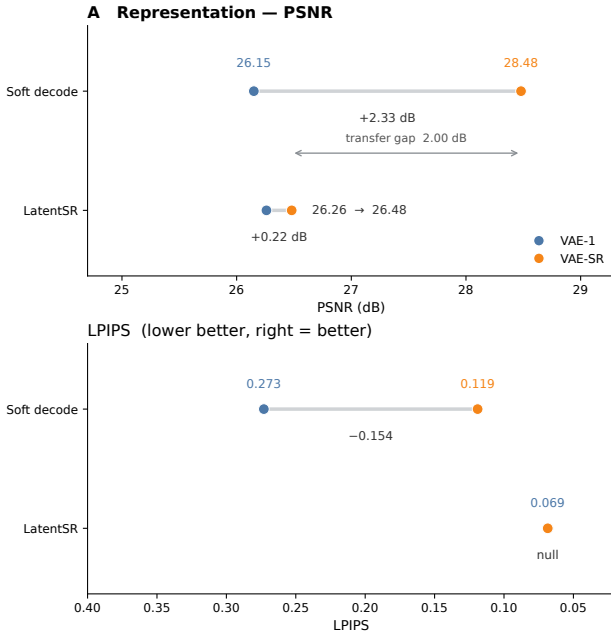


Figure 4: Representation vs. consumption, $n=64$. Soft-decode of z_{lr} gains +2.33 dB; concat LATENTSR gains +0.22 dB and LPIPS is a null (LPIPS axis reversed so better is right).

Table 6: Matched concat / FiLM SR, $n=64$. Soft-decode is $D(z_{lr})$.

	soft PSNR	SR PSNR	soft LPIPS	SR LPIPS
VAE-1 concat	26.15	26.26	0.273	0.0683
VAE-SR concat	28.48	26.48	0.119	0.0685
VAE-SR FiLM	28.48	26.45	0.119	0.0696

Paired tests on matched indices (Table 7). The concat +0.22 dB is real and small; LPIPS is a null. Secondary SSIM (+0.0023) and Sobel edge MAE (−0.0028) move with PSNR. Imagewise Spearman ($n=64$): $\rho(\Delta\text{PSNR}_{\text{soft}}, \Delta\text{PSNR}_{\text{SR}}) = 0.049$, $\rho(\Delta\text{LPIPS}_{\text{soft}}, \Delta\text{LPIPS}_{\text{SR}}) = 0.053$, $\rho(\Delta\text{PSNR}_{\text{SR}}, \text{bicubic}) = -0.139$, $\rho(\Delta\text{PSNR}_{\text{SR}}, \text{HR Sobel}) = +0.182$. Images that gained the most in z_{lr} did not gain more after diffusion, so a λ_{align} sweep is not the next question. In latent space VAE-SR shifts $\text{RMSE}(z_{lr}, z_{hr})$ left while $\text{RMSE}(z_{sr}, z_{hr})$ overlaps VAE-1: the sampler lands in the same neighborhood despite a better condition.

5.2 Injection

Spatial FiLM on the same frozen VAE-SR does not close the gap (Table 6, last row; Table 7). Train loss at epoch 50: concat 0.1333 vs. FiLM 0.1343 (val 0.1342 vs. 0.1363)—same plateau, FiLM slightly worse. PSNR CI includes 0; LPIPS is not a perceptual win. Soft-decode is identical (same VAE). FiLM+VAE-1 was not trained (§4.4). The hole is not that the first convolution cannot see z_{lr} .

Table 7: Paired tests, $n=64$. Concat row: VAE-SR SR minus VAE-1 SR. FiLM row: FiLM minus concat, both on VAE-SR.

Contrast	Metric	Δ mean	95% CI	p
concat transfer	PSNR	+0.217	[+0.133, +0.300]	10^{-4}
concat transfer	LPIPS	+0.0002	[−0.0012, +0.0017]	0.76
FiLM vs. concat	PSNR	−0.032	[−0.089, +0.023]	0.28
FiLM vs. concat	LPIPS	+0.0011	[+0.0000, +0.0022]	0.060

5.3 Reverse chain

Table 8 and Figure 5 localize the miss. VAE-SR copies z_{lr} mid-reverse (cos = **0.985** at $t=656$); VAE-1 never does (peak 0.783, coincidentally VAE-SR’s final number). At $t=0$ most of the extra copy is gone (0.615 vs. 0.786). $\text{RMSE}(z_{lr}, z_{hr})$ is constant in t : the VAEs differ; what changes is the trajectory. That is why concat and FiLM can both see a better code and still emit nearly the same z_{sr} .

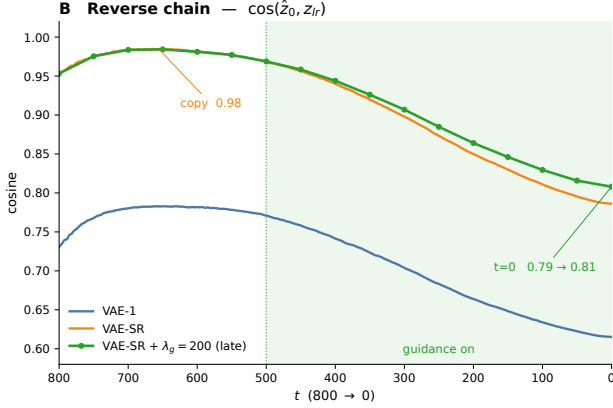


Figure 5: Reverse-chain cosine, $n=64$. VAE-SR copies z_{lr} near $t=656$ then the HR prior collapses the copy; late $\lambda_g=200$ ($t \leq 500$) slows that collapse.

Table 8: $\cos(\hat{z}_0(t), z_{lr})$ along reverse sampling, $n=64$.

	t	VAE-1	VAE-SR
	999 (noise)	0.282	0.543
	800	0.730	0.953
	656 (peak)	0.783	0.985
	500	0.771	0.969
	400	0.741	0.940
	0 (clean)	0.615	0.786

Latent cosine is not an image. Table 9 decodes $\hat{z}_0(t)$ vs. x_{hr} ; this is not soft-decode $D(z_{lr})$. Best absolute PSNR is near $t=400$ for both VAEs. The $t=0$ Δ is exactly the concat +0.22 dB. Soft-decode 28.48 is not $D(\hat{z}_0)$ at the copy ($t=650$: 26.70 dB).

Table 9: $D(\hat{z}_0(t))$ vs. x_{hr} , $n=64$. Not soft-decode.

t	PSNR VAE-1	PSNR VAE-SR	Δ
800	20.86	22.64	+1.78
650	26.19	26.70	+0.51
400	27.83	28.23	+0.39
0	26.26	26.48	+0.22

5.4 Guidance

Inference only: frozen VAE-SR + Q2 concat (§4.6). Every job reprints $\text{PSNR}(D(z_{lr}), x_{hr}) = 28.48$ on indices 0..63 or is discarded.

Stage 1 ($n=64$, $\lambda_g=200$): early $t \leq 800$ and late $t \leq 500$ both reach 27.60 dB (Table 10). All 64 images gain PSNR. LPIPS CIs include 0 (early $p=0.29$, late $p=0.39$). Trajectory moved: $t=0$ cosine 0.786 $\rightarrow$ 0.808, $\mathcal{L}_g$ 0.00128 $\rightarrow$ 0.00048—not a scale failure. The windows tie to 0.003 dB; extra steps $t \in (500, 800]$ lift mid-chain

$D(\hat{z}_0)$ then late catches up by $t \approx 450$. The finish comes from slowing the post-copy collapse ($t \leq 500$). Stage 2 is late-only; no combined schedule.

Table 10: Stage 1 windows, $n=64$, $\lambda_g=200$.

Method	PSNR	LPIPS
Soft-decode (VAE-SR)	28.48	0.119
Unguided LATENTSR	26.48	0.0685
Early $t \leq 800$	27.60	0.0695
Late $t \leq 500$	27.60	0.0693

Stage 2, same 64, late $t \leq 500$, $\lambda_g \in \{50, 200, 800\}$ (Table 11, Figure 6). Gap $28.482 - 26.478 = \mathbf{2.004}$ dB. $\lambda_g=50$: smaller recon gain, LPIPS better on $\approx 80\%$ of images. $\lambda_g=200$: largest PSNR with no LPIPS tax—operating point. $\lambda_g=800$: PSNR toward soft-decode, LPIPS toward 0.119 (over-guidance). $t=0$ cosine 0.786 $\rightarrow$ 0.794/0.808/0.837: guidance does not lock to the 0.985 copy peak.

Table 12 is the pre-registered confirmation (val_index 0..255, same checkpoints, late window). Do not mix it with Table 11. Bars §4.6 all pass: 255/256 images gain PSNR at every λ_g (sole miss: val_index 99, 162870.jpg); ΔLPIPS is monotone $50 < 200 < 800$; all PSNR CIs exclude 0. Soft-decode on 256 is 28.41 dB (first 64 still 28.48): extra images, not a swapped VAE. Gap $28.412 - 26.335 = \mathbf{2.078}$ dB; recovery 27%/56%/80% matches $n=64$. Unlike discovery, $\lambda_g=200$ LPIPS is significantly *better* than unguided ($p=0.0017$). Operating point remains late $\lambda_g=200$: 26.33 $\rightarrow$ 27.50 dB (56% of the gap), LPIPS not walking to 0.119.

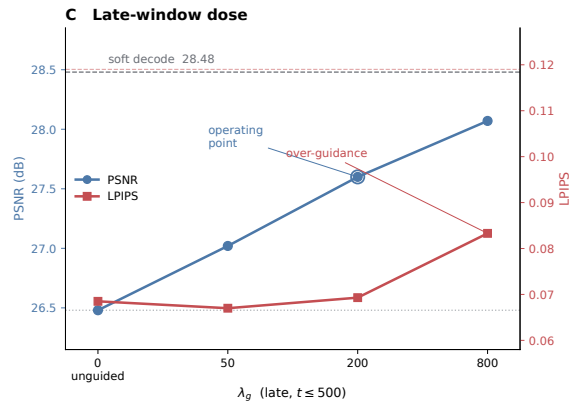


Figure 6: Late-window dose, $n=64$. $\lambda_g=200$ is the operating point; $\lambda_g=800$ walks LPIPS toward soft-decode 0.119. Confirmatory $n=256$: Table 12.

Table 11: Late-window dose, $n=64$ (discovery).

λ_g	PSNR	LPIPS	Δ PSNR	% of gap
0	26.48	0.0685	—	—
50	27.02	0.0670	+0.54	27%
200	27.60	0.0693	+1.12	56%
800	28.07	0.0833	+1.59	80%
soft-decode	28.48	0.119	+2.00	100%

Table 12: Confirmatory late-window dose, $n=256$. Δ vs. unguided with 95% CI.

Method	PSNR	LPIPS	Δ PSNR	% gap
Soft-decode	28.41	0.119	—	—
Unguided	26.33	0.0720	—	—
Late $\lambda_g=50$	26.90	0.0698	+0.57 [0.53, 0.60]	27%
Late $\lambda_g=200$	27.50	0.0707	+1.16 [1.11, 1.22]	56%
Late $\lambda_g=800$	28.00	0.0832	+1.66 [1.60, 1.72]	80%

6 Discussion

The representation hypothesis is half right. An SR-aware VAE really does improve z_{lr} : soft-decode gains +2.33 dB and LPIPS 0.273 $\rightarrow$ 0.119 while HR recon stays at 37.3 dB (Table 5). That improvement does not automatically become better SR. Under a matched concat DDPM the SR PSNR gain is +0.22 dB, LPIPS is a null, and imagewise Spearman $\rho(\Delta_{\text{soft}}, \Delta_{\text{SR}}) \approx 0.05$ (Table 6, Table 7). Spatial FiLM on the same frozen VAE-SR is also a PSNR null (Table 7). The missing link is not “the first convolution cannot read z_{lr} .”

The reverse-chain diagnostic is the more useful localization. $\hat{z}_0(t)$ under VAE-SR becomes strongly aligned with z_{lr} mid-sampling (cosine 0.985 at $t=656$) and then loses that alignment by $t=0$ (Table 8, Figure 5). Decoding $\hat{z}_0(t)$ is not soft-decode: absolute PSNR peaks near $t=400$, and the $t=0$ PSNR Δ matches the concat +0.22 dB (Table 9). We treat the late drop as an empirical fact about this reverse process, not as a proof that an isolated “HR prior” is the cause.

Guidance tests that fact at inference, with no extra training. Early and late windows at $\lambda_g=200$ tie (Table 10): the finish is in $t \leq 500$. A late dose recovers 27%/56%/80% of the transfer gap at $\lambda_g \in \{50, 200, 800\}$; $\lambda_g=800$ walks LPIPS toward soft-decode 0.119 (Table 11, Figure 6). The confirmatory table is $n=256$ (Table 12): late $\lambda_g=200$ moves 26.33 $\rightarrow$ 27.50 dB (56% of the 2.08 dB gap) and slightly improves LPIPS (0.0720 $\rightarrow$ 0.0707). That is the operating point: recover a large share of the unused representation gain without collapsing to the soft-decode image.

What we do not claim. This is not a new SR back-

bone, tokenizer, or trained guidance network, and it is not a SOTA comparison against StableSR, SeeSR, or DiffBIR. The FiLM null is for this spatial conditioner on this frozen VAE-SR code; FiLM+VAE-1 was not trained, so we do not claim a VAE-conditioner interaction. Soft-decode $D(z_{lr})$ is a probe of the condition, not an SR method. We do not claim that autoencoder quality transfers to SR in general.

Limits. The protocol is $4\times$ bicubic CelebA 32 $\rightarrow$ 128 with a small latent UNet trained from scratch. VAE, concat, FiLM, and trajectory numbers are $n=64$; the guidance dose we treat as confirmatory is $n=256$. Guidance was not uniformly helpful: one image (val_index 99) had PSNR that worsened monotonically as λ_g increased, so the intervention is not failure-proof per image even when the mean is strongly positive (Table 12). Full validation ($\approx 19,867$ images) is not run. Guided sampling back-propagates through the frozen decoder at every guided step and costs ≈ 21 s/image in our protocol (§2.4), which is why the full split was infeasible. We did not sweep λ_{align} . $\text{Cosine}(z_{lr}, z_{hr})$ plateaued after epoch 1 of the single $\lambda_{\text{align}}=10^{-3}$ fine-tune (§4.2); that is a training-curve observation, not a hyperparameter study, so we do not claim 10^{-3} is optimal. $\lambda_g=200$ is a compromise on $R(t)$, not a grid search over SR PSNR. A claimed related paper could not be verified from the citation as given and was excluded (§2.5).

The practical implication is narrow and, we think, useful. If a better z_{lr} does not move the SR table, the next experiment is not necessarily a fancier injector. Measure the reverse chain; if alignment appears and then fades, inference-time decoder-side consistency on that window is a cheap test of whether the unused information can still be spent.

7 Conclusion

Better LR latents need not yield better latent diffusion SR. On a controlled CelebA $4\times$ protocol, an SR-aware VAE improves soft-decode by +2.33 dB, yet a matched concat DDPM—and spatial FiLM on the same frozen code—consume almost none of that gain. The reverse chain becomes strongly aligned with z_{lr} mid-sampling and then loses that alignment before $t=0$. Decoder-side LR-consistency guidance on the late window, with no extra training, recovers 56% of the 2.08 dB transfer gap at $\lambda_g=200$ ($n=256$: 26.33 $\rightarrow$ 27.50 dB) and slightly improves LPIPS. The contribution is that diagnosis and that inference-time test, not a new SR architecture.

Code availability

Training and evaluation code, configs, and the experiment log are at <https://github.com/HusseinHanafy207/LatentSR>. Paired checkpoints used in this report are at <https://huggingface.co/HusseinHamouda/LatentSR-checkpoints>.

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